

General Chemistry — Midterm Exam — Answer Key

Set B

1. Covalent bonds involve the sharing of ? **Answer: electrons**
Shared electron pairs.
2. What is the atomic number of carbon? **Answer: 6**
Carbon has 6 protons.
3. The pH of a neutral solution is? **Answer: 7**
Neutral pH at 25 C.
4. Balance: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ **Answer: A**
 $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}.$
5. Which gas law relates pressure and volume at constant temperature? **Answer: B**
Boyle's law: $P_1V_1 = P_2V_2$.
6. Design an experiment to determine the molarity of an unknown acid. **Answer: Titrate with a base of known concentration using an indicator; calculate molarity from the equivalence point.**
Titration procedure.
7. Avogadro's number is 6.022×10^{23} . **Answer: True**
Particles per mole.